PRISM: Predictive, Explainable System-Health Monitoring on a Personal Computer, with a Three-Model Comparison of Resource Forecasters

Sahil Dhanuka Shambhav Ghosh Naresh Kumar Bishnoi

***Abstract*—Desktop operating systems report resource usage only after a slowdown has already happened. We present PRISM, a monitor for a personal computer that runs entirely on the device. It forecasts CPU and memory utilisation 45 s and 210 s ahead, estimates the probability of resource contention, explains the likely cause, and recommends an action. PRISM has 592 automated tests, a browser dashboard and a measured overhead of 2.26 % CPU. To choose its forecasting model, we compared three machine-learning regressors, LightGBM, Random Forest and Ridge regression, on 113,876 samples (about 48 h) of real usage from one macOS machine. We first added a preprocessing stage fitted on training data only. It removes rows whose history is incomplete after collector gaps, log-transforms skewed byte rates, and drops constant and near-duplicate features (254 to 224). All three models then used identical inputs and chronological splits, under a selection rule fixed in advance. LightGBM had the highest mean skill against a naive persistence forecast (-0.113, versus -0.252 for Random Forest and -0.607 for Ridge) and the smallest worst-case loss (23 % versus 85 % and 179 %), and it was the cheapest to run. It was therefore selected. The evidence is mixed, and we report it as such. Preprocessing lifted Ridge's mean skill from -0.98 to -0.61 and made it the best CPU forecaster; it is the only model that reliably beats persistence (+12 % at 45 s). Random Forest had the best cross-validation record. A regime analysis explains the negative average skill. On the 25 % of samples where CPU load actually changes, the learned models cut error by 29–34 % against persistence. On the 75 % where nothing changes, persistence is almost unbeatable, and those samples dominate this idle-heavy machine. Expressed as accuracy (100 % minus mean absolute error), LightGBM's forecasts are 97.9 % accurate on average, 96–97 % for CPU and 99 % for memory.**

***Index Terms*—system monitoring, time-series forecasting, LightGBM, random forest, ridge regression, resource contention, explainable AI, walk-forward validation, Diebold–Mariano test**

I. Introduction

Operating-system tools such as Activity Monitor on macOS and Task Manager on Windows are *reactive*. They show that the CPU is saturated or that memory is swapping only after the user has already felt the slowdown. They also report raw numbers, not causes. A user who sees "memory pressure: high" still has to work out which process is responsible, whether the pressure is growing, and what to do about it. Most users never open these tools at all until the machine has already become unresponsive.

PRISM aims to make desktop monitoring *predictive* and *explainable*. Every few seconds it answers four questions:

1) Where will CPU and memory utilisation be in the next 30–60 s (short horizon) and 2–5 min (medium horizon)?
2) How likely is resource contention in the short horizon?
3) If contention is likely, what kind is it? Examples are a single runaway process, several competing processes, or gradual versus sudden memory pressure.
4) What should the user do about it?

Answering the first question well is the foundation for the others, because the forecasts are inputs to the contention classifier and to the diagnosis. The choice of forecasting model is therefore the most important modelling decision in the system. We study it through three research questions.

RQ1 Which of three standard ML regressors (boosted trees, bagged trees, a regularised linear model) forecasts desktop CPU and memory utilisation best on real usage data, and are the differences statistically significant?

RQ2 Why do the learned models struggle to beat a naive "no change" forecast, and where, if anywhere, do they add value?

RQ3 Which features does the chosen model actually rely on?

This paper makes four contributions:

- **A complete on-device pipeline**, from threaded metric collection to a browser dashboard. The same feature-engineering code is used for training and live inference. The pipeline has an explicit deployment gate and human-confirmed retraining (Section IV).
- **A leakage-free preprocessing stage and a controlled comparison of three ML forecasters** on real usage data, run with and without preprocessing, with a selection rule fixed in advance (Sections V and VI).
- **A statistical and diagnostic analysis** that goes beyond average error. It includes Diebold–Mariano tests, block-bootstrap confidence intervals, a stable-versus-moving regime analysis and feature attribution.
- **An honest account of the limits** of the result, and of how the system's own gate stops a weak model from being deployed (Section VII).

II. Related Work

A. Workload and resource forecasting

Most resource prediction work targets servers and clouds, where forecasts drive autoscaling. PRESS [1] used signal

TABLE I
PRISM COMPARED WITH EXISTING TOOLS AND WORK

	Activity Monitor / Task Mgr.	PRESS [1]	ARIMA [2], [3]	**PRISM**
Target	PC	cloud	cloud	**PC**
Predicts load	no	yes	yes	**yes**
Method	–	FFT + Markov	ARIMA	**LightGBM**
Explains cause	no	no	no	**SHAP**
Advises user	no	auto-scales	auto-scales	**3 actions**
Flags anomalies	no	no	no	**Isol. Forest**
Trains on device	–	no	no	**yes, gated**

processing and Markov chains to predict resource demand for elastic scaling. Roy *et al.* [2] and Calheiros *et al.* [3] used ARIMA-family models for workload forecasting and measured their effect on quality of service. These systems see aggregated, fairly smooth demand from many users. PRISM differs in two ways. First, it targets a single interactive personal computer, where load is dominated by long idle periods broken by short, user-driven bursts. Second, it runs both training and inference on that same machine under a strict overhead budget, so model cost is a first-class concern. Table I summarises the differences, including from the built-in tools PRISM aims to improve on.

B. Model families for tabular forecasting

Gradient boosting [4] builds an additive model of shallow trees, each fitted to the residuals of the ensemble so far. XGBoost [5] and LightGBM [6] made the approach fast and scalable. LightGBM uses histogram-based split finding and grows trees leaf-wise. Random Forests [7] instead average many deep, decorrelated trees, which reduces variance. Ridge regression [8] is the standard regularised linear baseline. On medium-sized tabular data, tree ensembles remain very hard to beat, even for deep networks [9]. We therefore compare the two main tree-ensemble families against a linear model rather than against deep sequence models, which would also be too expensive to retrain on the user's own machine.

C. Anomaly detection and explainability

Anomaly detection is a broad field [10]. Isolation Forest [11] is popular for unsupervised scoring because it is fast and needs no labels. SHAP [12] gives consistent per-prediction feature attributions, and TreeSHAP [13] computes them exactly and quickly for tree ensembles. This is one practical reason to prefer a tree model when predictions must be explained to a user.

D. Evaluating forecasts

Random k-fold cross-validation leaks future information into training for time series. Rolling-origin (walk-forward) evaluation is the accepted alternative [14], [15]. Scale-free comparison against a naive forecast [16], [17] is the usual way to judge whether a model has any real skill. Whether one forecaster is significantly more accurate than another is tested with the Diebold–Mariano (DM) test [18], using a variance estimate that is robust to autocorrelation [19]. For confidence intervals on dependent data, the moving-block bootstrap [20] resamples contiguous blocks, not individual points. We use all of these.

III. Problem Formulation

Let $\mathbf{m}_t$ be the vector of system metrics sampled at time t. These are CPU %, memory %, swap %, one-minute load average, and disk and network byte rates. Let $\mathbf{p}_t$ be a summary of the process table at t. A feature map ϕ turns the recent history into a fixed-length vector $\mathbf{x}_t = \phi(\mathbf{m}_{\le t}, \mathbf{p}_{\le t}) \in \mathbb{R}^{254}$, using only data at or before t.

A. Forecasting

For a metric $y \in \{\text{CPU}\,\%, \text{memory}\,\%\}$ and a horizon $h \in \{45\,\text{s}, 210\,\text{s}\}$ (the midpoints of the two horizon ranges), we learn a function f that predicts the *change*, not the level:

$$\hat{y}_{t+h} = y_t + f(\mathbf{x}_t), \qquad f = \arg\min_f \sum_t \ell\big(y_{t+h} - y_t,\ f(\mathbf{x}_t)\big). \tag{1}$$

Predicting the delta makes "no change" the natural zero of the model. It also removes the slowly varying level, which a tree model cannot extrapolate beyond its training range. Four such models are trained: two metrics times two horizons.

B. Contention

A sample is labelled as contention if any tracked metric k reaches its threshold τ_k at the short horizon:

$$c_t = \mathbb{1}\Big[\max_k \big(m_{k,t+h_s} - \tau_k\big) \ge 0\Big]. \tag{2}$$

The thresholds are CPU 70 %, memory 85 %, swap 50 % and 50 kB/s for each disk and network byte rate. A classifier estimates $P(c_t = 1 \mid \mathbf{x}_t)$.

C. Evaluation measures

For n holdout samples with error $e_t = y_{t+h} - \hat{y}_{t+h}$, we report MAE $= \frac{1}{n}\sum |e_t|$ and RMSE $= \sqrt{\frac{1}{n}\sum e_t^2}$ in percentage points (pp). *Directional accuracy* is the fraction of samples where the predicted direction (rising, falling or stable) matches the actual one. A change within 2 % of the current value (relative) counts as stable. The primary measure is *skill* against the naive persistence forecast $\hat{y}_{t+h} = y_t$:

$$S = 1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{persistence}}}. \tag{3}$$

$S > 0$ means the model beats "no change", $S = 0$ means it ties, and $S < 0$ means it is worse. For example, $S = -0.18$ means 18 % more error than persistence.

IV. System Design

Fig. 1 shows the pipeline. It is implemented in Python 3.12. The main libraries are `psutil` 7.2 for metrics, SQLAlchemy 2.0 over SQLite, LightGBM 4.7, scikit-learn 1.9, statsmodels 0.14, SHAP 0.52 and FastAPI 0.141. The dashboard is plain JavaScript with uPlot charts and needs no build step.

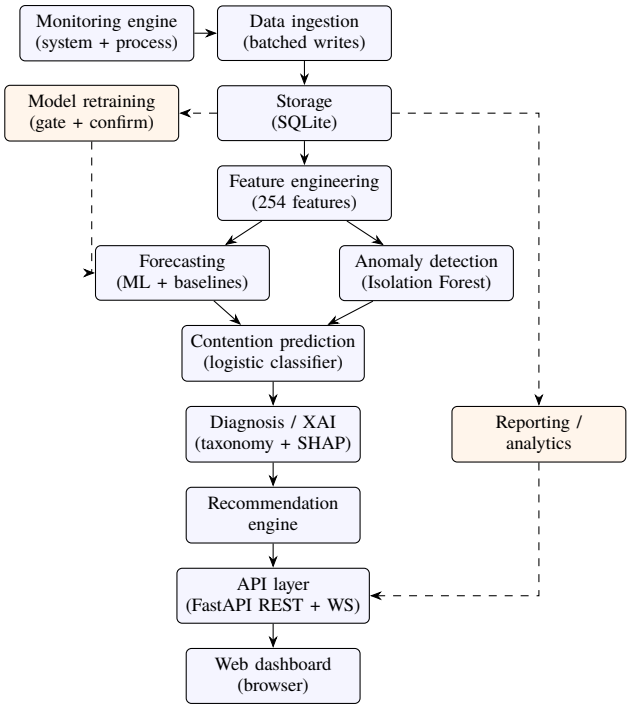


Fig. 1. PRISM's pipeline. Solid arrows show the live inference cycle, which runs every 5 s. Dashed arrows show loops that run on their own cadence: retraining on accumulated real usage, and reporting.

TABLE II
FEATURE GROUPS (254 COLUMNS IN TOTAL)

Group	Cols	Examples
Current value	1	metric value at t
Lagged values	16	value 1–k samples ago
Rate of change	8	per-second slope of each gauge
Rolling-window	64	mean, std, min, max over 30 s and 120 s
Process activity	165	launches in 30/120 s, terminations in 30 s, launch bursts in 10 s, activity of processes younger than 60 s, top-1/top-5 CPU and RSS shares, thread counts (and their rolling statistics)

Algorithm 1 One live-inference cycle (every 5 s)

1: $\mathbf{x}_t \leftarrow \phi$(recent metrics and process summaries)
2: **for** each metric y and horizon h **do**
3: $\quad \hat{y}_{t+h} \leftarrow y_t + f_{y,h}(\mathbf{x}_t)$ ▷ active model
4: $\quad c_{y,h} \leftarrow \text{clip}(1 - \text{RMSE}_{\text{val}}/\sigma_y,\, 0,\, 1)$
5: **end for**
6: $a_t \leftarrow$ IsolationForest score of $\mathbf{x}_t$ in $[0, 1]$
7: $p_t \leftarrow P(c_t{=}1 \mid \mathbf{x}_t, \hat{\mathbf{y}}) \cdot (1 - 0.3\, a_t)$
8: **if** $p_t \geq \tau$ **then**
9: $\quad$ category $\leftarrow$ DIAGNOSE(SHAP top-5, top processes)
10: $\quad$ actions $\leftarrow$ CATALOG(category)
11: **end if**
12: store all outputs; push state to WebSocket clients

A. Data collection and storage

Background threads poll system metrics every 1.5 s. They poll the process table every 3 s, storing the top 20 processes by CPU and the top 20 by memory individually and aggregating the rest into one row. A typical desktop has around 180 processes, so this cuts storage by an order of magnitude. A bounded queue of 20,000 items feeds batched, ordered writes of 200 rows or every 5 s, whichever comes first. Retention is 72 h at full resolution for system metrics, after which rows are rolled up, and 6 h for raw process rows. Durable per-poll process summaries are computed and stored *before* the raw rows are pruned, so multi-day training data keeps its process signal.

B. Feature engineering

One pure function builds the feature matrix for both training and live inference. This removes train/serve skew by construction, and a dedicated test checks that a live feature vector equals the training vector for the same timestamp. A second test checks causality: no feature uses data from after t. Table II summarises the 254 features. A separate preprocessing stage (Section V-B) then cleans the feature matrix before any model sees it.

C. Live inference loop

Algorithm 1 shows one inference cycle, which runs every 5 s. This is deliberately slower than the 1.5 s polling rate, because each cycle (especially a SHAP explanation) costs more than one poll, and the 3 % CPU budget covers inference as well as collection.

The confidence $c_{y,h}$ in line 4 is a single number between 0 and 1. It is 1 for a perfect validation fit and 0 when the validation RMSE is as large as the target's own standard deviation. It is not a calibrated interval. Line 7 *damps* the contention probability when the current state looks anomalous. The classifier was trained on normal patterns, so its output is less trustworthy on unusual inputs. The decision threshold τ is fitted on held-out data as the point of the precision–recall curve with the highest F1 score; for the deployed classifier it is 0.50. When the held-out labels contain only one class, the curve is unusable and τ falls back to 0.75, which favours precision over recall: a false alarm costs user trust, while a missed alarm costs only a warning that the user would not have had anyway.

D. Diagnosis and recommendation

When contention is predicted, the top five attributed features are mapped to coarse resource types (CPU, memory, disk, network). If no type accounts for at least 60 % of the total attribution, the event is *cross-resource* contention. Otherwise the dominant type chooses the category, refined by further rules. Table III lists the eight categories. Each actionable category maps to three curated actions (21 in total) from a versioned catalogue. The anomaly category is deliberately not actionable, because recommending an action for a pattern the system does not understand would be guesswork. Every diagnosis has a technical explanation and a plain-language one.

TABLE III
DIAGNOSIS TAXONOMY AND AN EXAMPLE RECOMMENDED ACTION

Category	Rule	Example action
CPU, single process	top process $\geq$50 % of top-5 CPU	Quit or pause the dominant process
CPU, multi-process	CPU dominant, no single leader	Stagger simultaneous heavy tasks
Memory, gradual	mem. rate $<$0.2 pp/s	Close unused tabs and idle apps
Memory, sudden	mem. rate $\geq$0.2 pp/s	Identify what just launched
Disk I/O bottleneck	disk dominant	Let the current disk job finish
Network saturation	network dominant	Pause large downloads
Cross-resource	no type $\geq$60 %	Reduce concurrent workload
Unusual pattern	anomaly score high	(none: not actionable)

E. Deployment gate and retraining

Three statistical baselines are always trained alongside the ML forecaster: persistence, a moving average and Holt exponential smoothing. An ML forecaster is deployed only if its mean walk-forward MAE beats the *best* of them by at least 5 % (relative). Otherwise the best baseline serves predictions. A retraining candidate must also beat persistence by 5 % on the chronological holdout and exceed 55 % directional accuracy. The gate's decision and its evidence are always stored. Retraining never deploys automatically: an operator must confirm through the CLI, the API or the dashboard's Model view, and any deployment can be rolled back. This rule is why a weak model cannot silently reach the live dashboard.

F. API and user interface

A FastAPI server exposes 28 REST routes (health, readiness, live state, history, reporting, model comparison, retraining, confirm and rollback) and one WebSocket, `/ws/live`, that pushes each cycle's state. The same server serves a browser dashboard with five views. *Live* (Fig. 2) shows current usage and contention risk, live charts of actual versus forecast values with a confidence band, the contention probability against its threshold, the diagnosis in plain language, the recommended actions and the top contributing processes. *Accuracy* plots predicted against actual values over a chosen window. *Reports* gives a daily or weekly summary. *Compare* draws the saved results of the model comparison in Section VI on demand: accuracy, error, error relative to persistence and directional accuracy per target, the stable-versus-moving split, the cross-validation folds and running cost. *Model* shows the active version and is where an operator confirms or rolls back a retrained model. An earlier PySide6 desktop client is still included but is no longer the main interface. A record/replay tool plays back a real recorded pressure event into a separate database at any speed, for reproducible demonstrations.

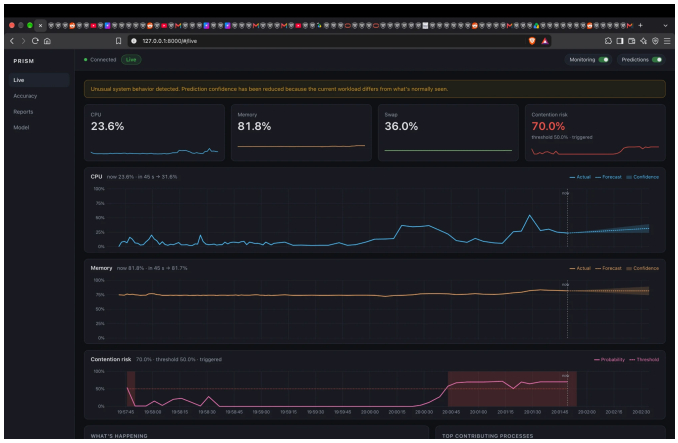


Fig. 2. The dashboard's Live view during a real memory-pressure event on the test machine. Contention risk has risen to 70 % against the fitted 50 % threshold, and the anomaly detector has flagged the workload as unusual, so prediction confidence is reduced. Below the charts (not shown) the diagnosis reads "Memory usage is climbing steadily" and the first recommended action is to close unused browser tabs. After the tabs were closed, memory fell from 82 % to 63 % and the risk returned to 0 %.

TABLE IV
PYTHON TESTS PER MODULE (B = BEHAVIOURAL, P = PLUMBING). THE DASHBOARD'S 24 JAVASCRIPT TESTS RUN SEPARATELY UNDER NODE.

Module	Tests	B	P
Data storage	31	14	17
Monitoring + ingestion	35	17	18
Feature engineering	45	29	16
Preprocessing	9	8	1
Forecasting	54	39	15
Contention + anomaly	76	48	28
Diagnosis / XAI	40	40	0
Recommendation	14	7	7
Model lifecycle (gate, deploy)	37	27	10
Model comparison + evaluation	8	3	5
API layer	63	16	47
Reporting / analytics	26	23	3
Web dashboard	6	2	4
Desktop client (legacy)	54	28	26
Cross-cutting (CLI, config, replay, …)	70	40	30
Total	**568**	**341**	**227**

G. Engineering quality

The system has 592 automated tests: 568 Python tests (Table IV) and 24 JavaScript unit tests for the dashboard's numeric helpers. Each Python test is tagged either *behavioural*, meaning it asserts that a computed value is correct, or *plumbing*, meaning it only asserts that code ran. The tag is enforced by the test runner, so a large test count cannot hide a module that is never checked for correctness. Every module has at least one behavioural test. Monitoring overhead was measured in isolation at 2.26 % CPU against a 3 % budget, and database growth at 3.48 MB/h against a 200 MB/h ceiling.

V. EXPERIMENTAL SETUP

A. Data

The data is real, passively collected usage of one macOS development machine between 17 and 23 September 2026.

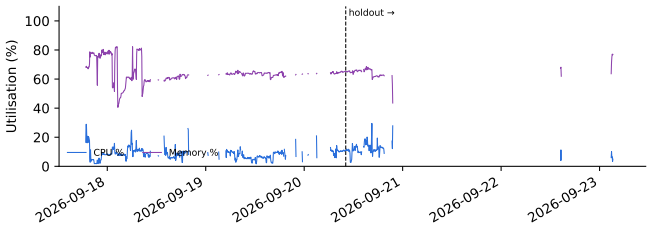


Fig. 3. CPU and memory utilisation over the collection window (5-min means). Blank stretches are periods when the collector was off. The dashed line marks the start of the holdout.

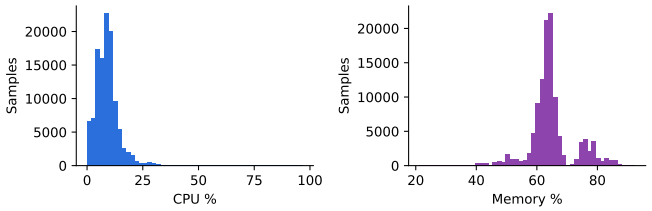


Fig. 4. Distribution of CPU and memory utilisation across all 113,876 samples. CPU is below 20 % in 96.7 % of samples.

The machine was used normally and no load was generated artificially. The wall-clock window is 130 h. The collector was off for two long stretches (41.0 h and 12.2 h) and paused 140 more times for periods of 30 s to a few minutes, mostly when the laptop slept. The data covers about 47.7 h of recorded time (113,876 samples at 1.5 s). The split is chronological with no shuffling. The oldest 80 % of samples (91,100) are used for training and the newest 20 % (22,776, from 20 September 10:07 UTC) are held out. To keep every number in this paper reproducible, all analyses run on a frozen copy of the database trimmed to this window.

Figs. 3 and 4 characterise the data, which is strongly idle-dominated. Median CPU is 8.8 %, the 90th percentile is 14.4 % and the 99th is 27.7 %. CPU is below 10 % in 65.0 % of samples and below 20 % in 96.7 %. It reaches the 70 % contention threshold in only nine samples (0.008 %). Memory is more stable still: the median is 64.0 %, 80 % of samples lie between 59.0 and 76.5 %, and 1.2 % are at or above 85 %. On a machine that is mostly idle, the best estimate of the next value is usually the current value, and this shapes every result below.

B. Data preprocessing

Feature engineering already performs the per-sample cleaning that any row needs: it sorts and de-duplicates timestamps, turns cumulative counters into per-second rates, treats a counter reset or a gap above 30 s as missing, and leaves a lag feature blank rather than reaching across a gap. We audited the raw data before modelling. It contains no duplicate timestamps, no out-of-order samples, no infinite values and no percentages outside 0–100 %. It does contain four problems that per-sample cleaning cannot fix, because they depend on the dataset as a whole:

TABLE V
PREPROCESSING STEPS AND THEIR EFFECT (45 S TARGETS; 210 S IN PARENTHESES WHERE DIFFERENT)

Step	Rule	Effect
Rows (stateless)		
1. Validity	drop non-finite values or values outside 0–100 %	0 rows
2. Warm-up	drop the first 120 s after each gap >30 s	581 (578) train, 407 (320) test
Columns (fitted on training rows)		
3. Log transform	$\mathrm{sign}(x)\log(1+\lvert x\rvert)$ on byte-rate features	44 columns
4. Constant	drop zero-variance columns	1 column
5. Redundant	drop the later of any pair with $\lvert r\rvert > 0.99$	29 columns
Result		254 → 224 features 90,313 → 89,732 train rows 22,557 → 22,150 test rows

- *Incomplete history after gaps.* A "120 s rolling mean" computed 3 s after the collector restarts is really a mean of two samples.
- *Near-duplicate features.* 29 feature pairs are correlated above 0.99, for example a lag and a rolling minimum of the same slowly changing process count.
- *A constant feature.* One process statistic never varies in the training data.
- *Heavy skew.* Disk and network byte rates reach skewness of 65: a download is orders of magnitude above idle traffic.

We therefore added a preprocessing stage between feature engineering and the models (Table V). It is implemented as a tested module of the system (`features/preprocessing.py`, 9 tests). It has two kinds of step. *Row* steps are stateless and applied to training and test rows alike. *Column* steps are *fitted*: which columns to transform or drop is decided from training rows only, then applied unchanged to validation and test rows. In walk-forward cross-validation the column steps are refitted inside every fold, so no information from any evaluation slice leaks into the choice. Missing values are deliberately left as blanks for the tree models, which split on them natively. Ridge fills them with the training median and standardises every feature inside its own pipeline, again fitted on training rows only.

The row counts in Table V are slightly below the raw sample counts. The supervised dataset already excludes samples that have no measured value exactly one horizon later, which happens just before every gap. Many of the 140 short gaps are brief wake-ups from sleep, with only a few samples between two pauses. Those samples have no future value, so the pauses around them merge. The training timeline therefore splits into 14 contiguous segments at the 45 s horizon and 19 at 210 s, and warm-up rows are removed at the start of each.

TABLE VI
CANDIDATE MODELS AND CONFIGURATION

Model	Family	Configuration
LightGBM [6]	Gradient-boosted trees	80 rounds, 15 leaves, learning rate 0.1, $\geq$5 samples/leaf; NaN handled natively
Random Forest [7]	Bagged trees	100 trees, max depth 12, $\geq$5 samples/leaf, seed 0
Ridge [8]	Regularised linear	$\alpha = 1$; median imputation and standardisation
Persistence	*Reference*	$\hat{y}_{t+h} = y_t$ (not a candidate)

C. Candidates

Table VI lists the three candidates. LightGBM uses PRISM's production hyperparameters unchanged. The other two use common, sensible settings, implemented with scikit-learn [21]. No candidate was tuned on the test data. All three receive exactly the same preprocessed feature matrix and the same delta target.

D. Protocol

Each candidate is first evaluated with 4-fold walk-forward cross-validation (expanding window) on the training portion [14], [15]. In fold k, it trains on the first k blocks and is scored on block $k{+}1$. The preprocessing column steps are refitted on each fold's training blocks. The candidate is then refitted on all training rows and scored once on the untouched test rows. We record MAE, RMSE, directional accuracy, fit time, median single-row prediction latency (over 200 predictions) and serialised model size. Fit time and latency include the preprocessing step, because the live system would have to pay for it too.

E. Selection rule

The rule was written into the comparison script before its first run, so the choice could not be fitted to the numbers. The winner is the candidate with the highest mean holdout skill (Eq. 3) across the four targets. If two candidates are within 0.02 of each other, the one with lower prediction latency wins, because the forecaster runs every cycle.

F. Statistical analysis

Mean errors alone cannot tell whether a difference is real. We therefore use three further tools.

- **Diebold–Mariano test** [18]. It tests the per-sample loss difference $d_t = |e_t^A| - |e_t^B|$, with a Newey–West variance [19] whose lag equals the horizon in samples (30 for 45 s, 140 for 210 s). Consecutive errors overlap and are strongly autocorrelated, so an ordinary t-test would overstate significance.
- **Moving-block bootstrap** [20]. It gives 95 % confidence intervals for each skill score, using 1,000 resamples of 400-sample (10-minute) blocks.

TABLE VII
HOLDOUT MAE IN PERCENTAGE POINTS (22,150 TEST ROWS AT 45 s, 21,733 AT 210 s). LOWER IS BETTER; **BOLD** MARKS THE BEST OF THE THREE CANDIDATES.

Target	LightGBM	Rand. Forest	Ridge	*Persist.*
CPU, 45 s	2.954	3.126	**2.662**	*3.017*
CPU, 210 s	3.702	**3.339**	3.365	*3.550*
Mem, 45 s	0.643	**0.632**	0.970	*0.536*
Mem, 210 s	**0.942**	1.415	2.131	*0.765*

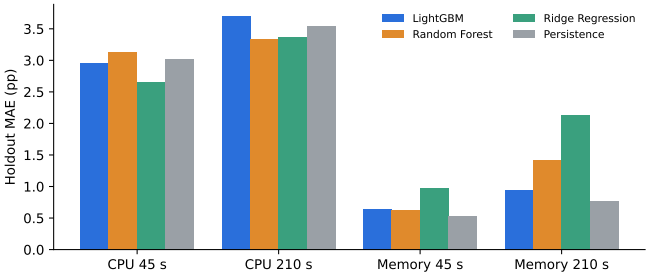


Fig. 5. Holdout MAE per target for the three candidates and the persistence reference.

- **Regime split**. Test rows are divided by the size of the actual change $|y_{t+h} - y_t|$. The *stable* regime is the smallest 75 % and the *moving* regime is the largest 25 %. The cut-offs are 3.8 pp and 5.0 pp for CPU at 45 s and 210 s, and 0.6 pp and 0.9 pp for memory. This split uses the realised outcome, so it is a *diagnostic* tool that shows where each model gains or loses. It is not a deployable rule.

VI. RESULTS

All results in this section use the preprocessed data unless stated otherwise. Section VI-D compares them with the same experiment run on raw features.

A. Holdout accuracy (RQ1)

Table VII and Figs. 5 and 6 give the holdout results. Each candidate is best on at least one target, and the pattern splits by metric.

- *CPU.* Ridge is best at 45 s (2.66 pp, 12 % below persistence) and close to the best at 210 s. Random Forest is best at 210 s. LightGBM beats persistence at 45 s by 2 % but not at 210 s.
- *Memory.* No model beats persistence. LightGBM is best at 210 s and within 0.011 pp of the best at 45 s. Random Forest and Ridge degrade badly at 210 s (85 % and 179 % worse than persistence).

Table VIII adds a second view. On *RMSE*, which penalises large errors more heavily, all three models *beat* persistence on both CPU targets; LightGBM's RMSE is 4.88 versus 5.48 at 45 s and 5.51 versus 6.02 at 210 s. The learned models make many small errors on idle samples and fewer large errors on bursts. Section VI-E confirms this directly. Directional accuracy on CPU is about 0.55 for all three models, against

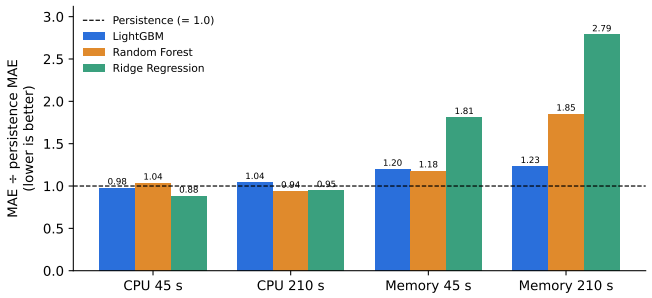


Fig. 6. Each model's holdout MAE divided by persistence's MAE. The dashed line at 1.0 is "no change". Bars above it are worse than persistence and bars below it are better. LightGBM stays within 1.23 of the line on every target; Random Forest reaches 1.85 and Ridge 2.79 on memory at 210 s. Ridge is below the line on both CPU targets.

TABLE VIII
HOLDOUT RMSE (PP) AND DIRECTIONAL ACCURACY

	RMSE				Directional acc.			
Target	LGB	RF	Rdg	Per.	LGB	RF	Rdg	Per.
CPU 45 s	4.88	4.94	**4.62**	5.48	.54	.54	**.57**	.10
CPU 210 s	5.51	**5.23**	5.24	6.02	.54	**.56**	.56	.09
Mem 45 s	1.30	1.38	1.54	**1.13**	.86	.89	.81	**.92**
Mem 210 s	1.67	3.37	2.99	**1.37**	.82	.81	.41	**.85**

0.10 for persistence. Persistence never predicts a change, so it is scored wrong whenever CPU moves by more than 2 %. On memory, which rarely moves that much, persistence's directional accuracy is naturally high.

For readers who expect a single percentage, Table IX restates the holdout MAE as accuracy, 100 − MAE, which is one minus the MAE normalised by the 0–100 % range. LightGBM's forecasts are 97.9 % accurate on average: about 97 % for CPU and 99 % for memory. The measure is generous on this data, because persistence reaches 98.0 % simply by repeating the current value. It is reported for readability only; the skill score of Section III, which compares each model with persistence directly, remains the basis for model selection.

B. Walk-forward stability

Table X and Fig. 7 show the cross-validation results. Here Random Forest has the lowest mean error on three of four targets and LightGBM on one. Ridge, despite its good CPU holdout, has the widest spread on three of the four targets, and its fold-1 CPU errors are 1.5–2.7 times those of the tree models. Every learned model improves steeply from fold 1 to fold 4 as it sees more data. On memory, the fold-1 errors are several times larger than the later ones, which is why the standard deviations are so wide. This suggests that the models are still data-limited.

C. Statistical significance

Tables XI and XII test how reliable the differences are.

- *LightGBM vs Random Forest.* LightGBM is significantly better on CPU 45 s ($p = 0.001$) and memory 210 s ($p = 0.005$). Random Forest is significantly better on CPU 210 s ($p < 0.001$). There is no significant difference on memory 45 s ($p = 0.61$). The head-to-head record is two wins, one loss and one tie, the same as without preprocessing.
- *LightGBM vs Ridge.* The record splits cleanly by metric. Ridge is significantly better on both CPU targets and LightGBM on both memory targets (all $p < 0.001$). LightGBM's memory advantage (0.33 and 1.19 pp) is larger than Ridge's CPU advantage (0.29 and 0.34 pp).
- *LightGBM vs persistence.* On CPU, LightGBM is statistically indistinguishable from persistence ($p = 0.35$ and 0.28), and both of its skill intervals contain zero. On memory it is significantly worse.

Only one model–target pair has a skill interval entirely above zero: Ridge on CPU 45 s, at $[0.07, 0.18]$. It is the only place in this study where a learned model beats persistence reliably, and it appears only after preprocessing.

TABLE IX
HOLDOUT ACCURACY IN PERCENT, 100 − MAE (THE TARGETS ARE THEMSELVES 0–100 %). PERSISTENCE SCORES AS WELL AS THE MODELS ON THIS IDLE-HEAVY TEST PERIOD.

Target	LightGBM	Rand. Forest	Ridge	*Persist.*
CPU, 45 s	97.0	96.9	**97.3**	*97.0*
CPU, 210 s	96.3	**96.7**	96.6	*96.5*
Mem, 45 s	99.4	**99.4**	99.0	*99.5*
Mem, 210 s	**99.1**	98.6	97.9	*99.2*
Mean	**97.9**	97.9	97.7	*98.0*

TABLE X
WALK-FORWARD CV MAE ON THE TRAINING PORTION, MEAN ± S.D. OVER 4 FOLDS.

Target	LightGBM	Rand. Forest	Ridge	*Persist.*
CPU, 45 s	2.98±0.89	**2.25±0.43**	3.28±1.65	*2.24±0.12*
CPU, 210 s	3.59±1.28	**3.34±0.96**	5.65±4.39	*2.58±0.29*
Mem, 45 s	1.89±2.30	**1.76±2.43**	2.31±2.39	*0.69±0.53*
Mem, 210 s	**3.28±4.14**	3.36±4.54	6.54±6.74	*1.14±1.06*

D. Effect of preprocessing

We ran the whole comparison twice, with and without preprocessing (Fig. 8, Table XIII). We compare skill rather than raw MAE, because removing warm-up rows also changes the test set, and skill measures each model against persistence on the same rows. The effect depends strongly on the model family.

- *Ridge gains the most*: its mean skill rises from -0.98 to -0.61, and both CPU targets turn positive. A linear model is sensitive to exactly the problems preprocessing fixes. Byte rates spanning six orders of magnitude dominate an unscaled linear fit, and 29 near-duplicate columns make its coefficients unstable. The log transform and redundancy filter remove both problems.
- *The tree models change little* (mean skill $-0.090 \rightarrow -0.113$ for LightGBM, $-0.245 \rightarrow -0.252$ for Random

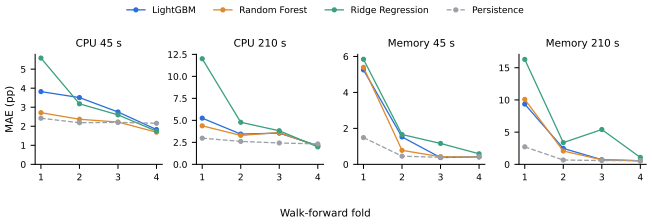


Fig. 7. Walk-forward MAE per fold. Errors fall sharply as each model sees more training data (fold 1 trains on the least data), and Ridge is the least stable.

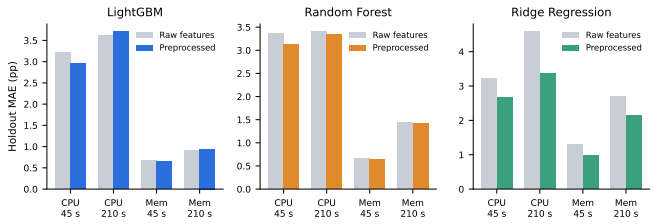


Fig. 8. Holdout MAE per model with raw features (grey) and after preprocessing (colour). Ridge improves on every target; the tree models change little.

TABLE XI
DIEBOLD–MARIANO TESTS ON ABSOLUTE ERROR, HOLDOUT. $\bar{d}$ IS LIGHTGBM'S MEAN ABSOLUTE ERROR MINUS THE RIVAL'S, IN PP; NEGATIVE MEANS LIGHTGBM IS BETTER. $^{*}p < 0.05$, $^{**}p < 0.01$.

Target	vs Rand. Forest $\bar{d}$	vs Rand. Forest p	vs Ridge $\bar{d}$	vs Ridge p	vs Persist. $\bar{d}$	vs Persist. p
CPU 45 s	−0.171	0.001**	+0.292	<.001**	−0.063	0.35
CPU 210 s	+0.363	<.001**	+0.337	<.001**	+0.153	0.28
Mem 45 s	+0.010	0.61	−0.328	<.001**	+0.107	<.001**
Mem 210 s	−0.473	0.005**	−1.189	<.001**	+0.178	<.001**

Forest). This is expected. A tree splits on thresholds, so a monotone transform such as the log leaves its possible splits unchanged, and trees tolerate redundant columns. The small shifts come from the warm-up rows and the dropped columns changing which splits are chosen. LightGBM gains on CPU 45 s and loses on memory 210 s.

Preprocessing is therefore not a free win for every model, but it is what makes a fair comparison possible. Without it, Ridge was being judged on inputs it cannot use.

E. Where the models win and lose (RQ2)

Fig. 9 answers RQ2. The negative average skill hides two very different situations.

- **Moving regime** (largest 25 % of changes). On CPU, LightGBM's MAE is 6.10 pp against persistence's 8.67 pp at 45 s (30 % lower) and 7.05 pp against 9.87 pp at 210 s (29 % lower). Ridge does slightly better (5.74 and 6.77, 34 % and 31 % lower) and Random Forest is similar (6.15 and 6.84). On memory, LightGBM ties with persistence (1.57 vs 1.58 at 45 s, and 2.18 vs 2.13 at 210 s), while Random Forest and Ridge are clearly worse at 210 s.
- **Stable regime** (smallest 75 %). Here persistence is excellent (1.13 pp on CPU at 45 s) and every model is worse (LightGBM 1.91, Ridge 1.64). The models occasionally predict a small change that does not happen, and that costs a little error on each of the many stable samples.

Because three quarters of the samples are stable, their small losses outweigh the large gains on the moving samples. This happens on MAE in particular, which weights every sample equally. It also explains why the learned models win on RMSE (Table VIII) but mostly lose on MAE. The learned models *have* learned something real about how CPU load changes.

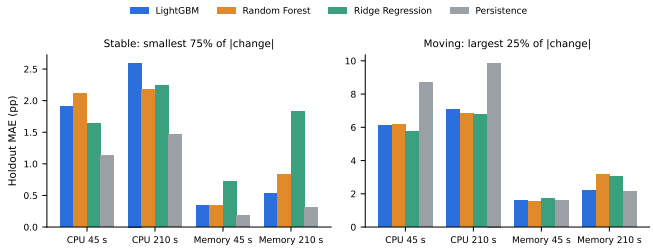


Fig. 9. Holdout MAE split by regime. Left: the 75 % of samples with the smallest actual change, where persistence is near-perfect. Right: the 25 % with the largest change, where every learned model cuts CPU error by about 30 % against persistence.

What they cannot yet do is recognise, before the fact, that the machine is in a stable period and stay quiet.

F. What the model relies on (RQ3)

Fig. 10 shows LightGBM's split-gain importance by feature group. For CPU, the single most important feature is the CPU rate of change: 49 % of total gain at 45 s and 34 % at 210 s. The current value comes next (14 % and 21 %). This fits the regime result, since the model mostly learns how the current trend continues. For memory, rolling-window statistics (29–33 %) and process-activity features (35–45 %) matter more. The largest single process features for memory at 210 s are the recent CPU use of newly started processes, the minimum thread count and a process-launch burst indicator. Process features make up 65 % of the kept columns (145 of 224) and receive 21–45 % of the gain, so each one is individually less important than a typical system feature. Gain importance shows what the model *uses*, not that a feature *improves* accuracy. We therefore do not claim that process features improve accuracy on this data.

G. Cost

Fig. 11 compares cost. Including preprocessing, LightGBM predicts one row in 1.74 ms, against 2.07 ms for Ridge and 15.8 ms for Random Forest (9× slower). The preprocessing transform itself adds about 1.4 ms per row for every model. The raw-feature run measured the models alone at 0.33 ms (LightGBM), 0.69 ms (Ridge) and 14.1 ms (Random Forest). LightGBM and Ridge fit in about 6 s, of which about 4.5 s is the correlation scan of the preprocessing step, while Random Forest takes 82–97 s. LightGBM models are about 138 kB,

TABLE XII
HOLDOUT SKILL S WITH 95 % MOVING-BLOCK BOOTSTRAP INTERVALS

Target	LightGBM	Random Forest	Ridge
CPU 45 s	0.02 [−0.04, 0.10]	−0.04 [−0.14, 0.05]	**0.12 [0.07, 0.18]**
CPU 210 s	−0.04 [−0.15, 0.05]	0.06 [−0.00, 0.12]	0.05 [−0.03, 0.12]
Mem 45 s	−0.20 [−0.29, −0.11]	−0.18 [−0.33, −0.04]	−0.81 [−0.98, −0.62]
Mem 210 s	−0.23 [−0.33, −0.12]	−0.85 [−1.41, −0.38]	−1.79 [−2.31, −1.28]

TABLE XIII
HOLDOUT SKILL S WITH RAW FEATURES → AFTER PREPROCESSING

Target	LightGBM	Random Forest	Ridge
CPU 45 s	−0.05 → +0.02	−0.10 → −0.04	−0.05 → **+0.12**
CPU 210 s	0.00 → −0.04	+0.06 → +0.06	−0.26 → **+0.05**
Mem 45 s	−0.18 → −0.20	−0.17 → −0.18	−1.29 → −0.81
Mem 210 s	−0.13 → −0.23	−0.78 → −0.85	−2.33 → −1.79
Mean	−0.090 → −0.113	−0.245 → −0.252	−0.984 → −0.607

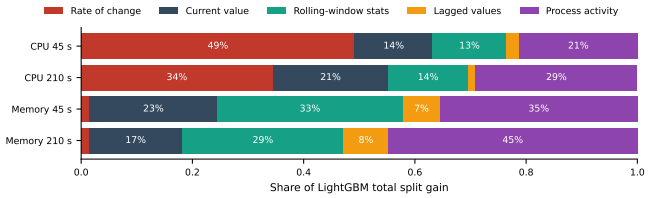


Fig. 10. Share of LightGBM's total split gain by feature group, per target.

Random Forest models 4.5–10.0 MB and Ridge models 17 kB. PRISM runs four forecasters every 5 s within a 3 % CPU budget and retrains on the user's own machine, so these differences matter in practice.

H. Choice and reasoning

Under the rule fixed in advance, **LightGBM is selected** (Table XIV). Its mean skill is 0.139 above Random Forest's and 0.494 above Ridge's, far outside the 0.02 tie-break band. The choice is the same as without preprocessing, but the evidence is now more mixed, and we report it as such. Ridge is the best CPU model, and Random Forest has the better cross-validation record. Four arguments support LightGBM as the single model PRISM ships.

1) *It has no bad target.* LightGBM is at most 23 % worse than persistence on any target. Random Forest is 85 % worse and Ridge 179 % worse on memory at 210 s, which is the horizon a user most needs for planning. A monitor that is occasionally very wrong damages trust more than one that is consistently close.
2) *Its wins are large and its losses small.* Against Random Forest it wins two targets significantly and loses one. Against Ridge its memory wins (0.33 and 1.19 pp) outweigh its CPU losses (0.29 and 0.34 pp).
3) *Cost.* It has the lowest latency and a fit time about 15× shorter than Random Forest's, which makes frequent on-device retraining practical.

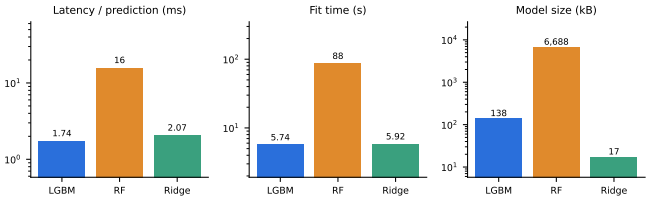


Fig. 11. Cost per model, averaged over the four targets (log scale). Latency and fit time include the preprocessing step.

TABLE XIV
SUMMARY USED FOR SELECTION (PREPROCESSED DATA)

	LightGBM	Rand. Forest	Ridge
Mean holdout skill S	**−0.113**	−0.252	−0.607
Worst target vs. persistence	**+23 %**	+85 %	+179 %
Best of 3, holdout targets	1/4	2/4	1/4
Best of 3, CV targets	1/4	**3/4**	0/4
Beats persistence reliably	–	–	CPU 45 s
LightGBM's DM record vs. rival	–	2W 1L 1T	2W 2L
Latency per prediction	**1.74 ms**	15.8 ms	2.07 ms
Fit time per target	≈6 s	≈88 s	≈6 s
Model size	138 kB	4.5–10 MB	**17 kB**

4) *Fit with the rest of the system.* It handles the missing lag features that follow collector gaps natively, supports exact TreeSHAP for the diagnosis module, and produces small models that are cheap to version and roll back.

The results also suggest a better design than any single model: Ridge for CPU and LightGBM for memory. We do not adopt it here. Choosing a model per target after seeing the test results would break the rule we fixed in advance, so it has to be confirmed on new data first (Section VII).

I. Comparison with statistical baselines and the gate

Table XV shows the separate harness that feeds the deployment gate. The moving average has the lowest error on all four targets, so the gate (Section IV-E) currently serves the moving average, not LightGBM. This agrees with everything above: on idle-dominated data, a model that smooths the recent past is hard to beat on average error. Fig. 12 shows the cost of that choice in the live system. The served forecast follows a sharp memory rise only after a delay, which is exactly the moving-regime weakness that the learned models address.

J. Contention classifier and overhead

The contention label is rare. In a 113,882-sample window, 1.5 % of samples are positive, and all of them fall in the train-

TABLE XV
WALK-FORWARD MAE OF LIGHTGBM VERSUS THE STATISTICAL BASELINES (EARLIER EVALUATION RUN ON RAW FEATURES, SAME DATA WINDOW, 113,882 SAMPLES).

Target	LightGBM	Persist.	Mov. avg.	Holt ETS
CPU, 45 s	3.15	2.13	**1.81**	2.87
CPU, 210 s	3.34	2.50	**2.19**	7.55
Mem, 45 s	2.85	1.07	**1.01**	1.66
Mem, 210 s	3.19	1.75	**1.68**	4.90

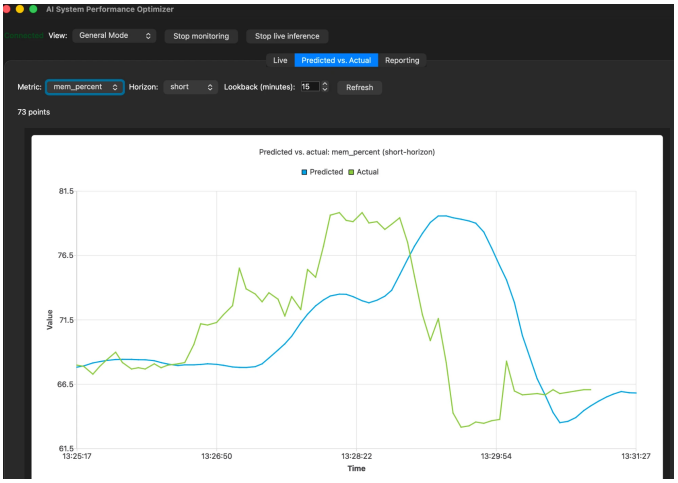


Fig. 12. Predicted-vs-Actual tab (memory, short horizon). The served, gate-selected baseline forecast (blue) lags a sharp rise and fall in actual memory use (green) by roughly the forecast horizon.

ing portion, so the chronological holdout contains *no* positives. Holdout precision, recall and F1 are therefore undefined, and we report them as undefined rather than as a score. A classifier trained on a genuine recorded pressure event (swap up to 91.8 %, CPU up to 78.7 %) is deployed and drives the live contention strip in Fig. 2. Its accuracy will be measured once a second, independent event falls inside a holdout window. Overheads are within budget: 2.26 % CPU and 3.48 MB/h of database growth.

VII. DISCUSSION

A. Why persistence is so hard to beat

Persistence is close to the best possible predictor whenever a series changes slowly relative to the forecast horizon. On this machine, CPU is below 20 % in 96.7 % of samples, and 80 % of memory samples lie within a 17.5-point band. Over 45 s, the typical change is smaller than the noise. A learned model has to pay for its ability to predict change: it sometimes predicts a change that does not happen. The regime analysis shows that this price is small per sample but applies to the majority of samples. Negative average skill on this data therefore reflects the workload, not a bug. The same effect is well known in forecasting generally, where naive benchmarks are routinely hard to beat [17].

B. What preprocessing taught us

Preprocessing changed which model is best on CPU at 45 s, from a near-tie between LightGBM and Ridge (3.21 vs 3.21 pp) to a clear, statistically reliable Ridge win. It did not change the winner, because the tree models are largely insensitive to it. Two lessons follow. First, a model comparison is only fair when each model gets inputs it can use; comparing a linear model on raw, skewed, redundant features understated it badly. Second, the fitted steps must be refitted inside every cross-validation fold. Fitting them once on all the data would let the choice of columns see the test period.

C. Implications for design

The results point to two next steps. The first is a *per-target* forecaster (Ridge for CPU, LightGBM for memory), to be confirmed on new data. The second is a *regime-aware* forecaster. A lightweight gate would predict whether the next window will be stable, using the rate-of-change and rolling-variance features that already dominate LightGBM's importance. It would serve persistence in stable periods and a learned model during transitions. Recovering even part of the moving-regime gain (about 30 % on CPU) without the stable-regime loss would give positive skill and let the deployment gate promote an ML model on its own evidence. The MAE-versus-RMSE split also suggests that the metric should follow the product goal: a monitor exists to catch bursts, which is closer to what RMSE rewards.

D. Threats to validity

1) *One machine, one user, two days of recorded data.* The results may not generalise to other workloads, and the fold-to-fold trend in Fig. 7 suggests that the models are still data-limited.
2) *No tuning.* LightGBM uses production defaults and the others use common defaults. Tuning could change the ordering.
3) *Preprocessing is not yet in the live app.* It is used in the comparison experiments and is a tested module, but the deployed forecaster does not call it yet.
4) *The raw and preprocessed runs use slightly different test rows*, because warm-up rows are removed. We compare them by skill, which is measured against persistence on the same rows.
5) *Different snapshot for Table XV.* It comes from an earlier run on raw features with its own training code, so only the ranking within it is comparable with the other tables.
6) *The regime split conditions on the outcome*, and *gain importance is not causality*. Both explain the results; neither is a deployable rule.

VIII. CONCLUSION AND FUTURE WORK

PRISM is a complete, tested, low-overhead pipeline for predictive and explainable desktop monitoring. We added a preprocessing stage that removes warm-up rows after gaps, log-transforms skewed byte rates and drops constant and redundant features. All learned steps are fitted on training

data only. With it, we compared three ML forecasters under a selection rule fixed in advance. LightGBM had the best mean skill and the smallest worst case, won its head-to-head against Random Forest, and was the cheapest to run. It is therefore PRISM's forecasting model. Preprocessing turned Ridge into the best CPU forecaster, the only model that reliably beat persistence anywhere. We have also stated clearly what the data does not yet show: on idle-dominated usage, no learned forecaster beats persistence on average, and the deployment gate correctly keeps a moving-average baseline in service. The regime analysis shows why, and it shows that the learned models add value exactly where a monitor needs it, when load is changing.

Future work, in order of expected impact:

1) a regime-aware or per-target forecaster (persistence when stable, Ridge for CPU, LightGBM for memory), confirmed on new data;
2) building the preprocessing stage into the live app and the deployed models;
3) longer and more varied collection across several machines and users, so that contention events fall in the holdout and the classifier can be graded;
4) hyperparameter tuning of all candidates under the same protocol;
5) calibrated prediction intervals, and verification on Windows.

Reproducibility

- Three-model comparison: `python scripts/compare_models.py` (`--raw` for the unprocessed run).
- Significance, bootstrap, regime and importance analysis: `python scripts/paper_analysis.py` (on the frozen snapshot; reproduces every holdout MAE in Table VII exactly).
- Figures: `python scripts/paper_figures.py`.
- Preprocessing module and tests: `features/preprocessing.py`, `tests/test_features_preprocessing.py`.
- Statistical-baseline table: `aiperfopt evaluate --report`.

References

[1] Z. Gong, X. Gu, and J. Wilkes, "PRESS: PRedictive Elastic reSource Scaling for cloud systems," in *Proc. Int. Conf. Network and Service Management (CNSM)*, 2010, pp. 9–16.

[2] N. Roy, A. Dubey, and A. Gokhale, "Efficient autoscaling in the cloud using predictive models for workload forecasting," in *Proc. IEEE Int. Conf. Cloud Computing (CLOUD)*, 2011, pp. 500–507.

[3] R. N. Calheiros, E. Masoumi, R. Ranjan, and R. Buyya, "Workload prediction using ARIMA model and its impact on cloud applications' QoS," *IEEE Trans. Cloud Computing*, vol. 3, no. 4, pp. 449–458, 2015.

[4] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[6] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[7] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[8] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[9] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.

[10] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.

[11] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, 2008, pp. 413–422.

[12] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[13] S. M. Lundberg *et al.*, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[14] L. J. Tashman, "Out-of-sample tests of forecasting accuracy: An analysis and review," *Int. J. Forecasting*, vol. 16, no. 4, pp. 437–450, 2000.

[15] C. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," *Information Sciences*, vol. 191, pp. 192–213, 2012.

[16] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," *Int. J. Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.

[17] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.

[18] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *J. Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.

[19] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.

[20] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[21] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.